

Mejdi Haddad

AI Engineer

RAG & LLM Agents | Data Science & ML | FastAPI & React

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SUMMARY

AI Engineer with a computer engineering background, building LLM applications, hybrid RAG systems, and ML solutions. Experienced with LangGraph workflows, evaluation and observability, Python/FastAPI backends, and React interfaces.

EXPERIENCE

Freelance AI Engineer | FullMetrix Aug 2025 – Present

- Built an AI assistant for a multi-tenant analytics platform: a LangGraph agent routes each question to SQL or document retrieval and answers with citations.
- Developed the AI assistant backend in Python with FastAPI and PostgreSQL, including the LangGraph orchestration layer.
- Used LangSmith for tracing and monitoring, with an evaluation harness to assess answer quality using representative merchant questions.

Generative AI Intern | Innomatics Research Labs Oct 2024 – Dec 2024

- Built a Streamlit/Gemini code reviewer that identifies Python bugs and suggests corrected code.
- Built an accessibility assistant with OCR, scene descriptions, and text-to-speech (OpenAI/Gemini, LangChain, Streamlit).
- Analyzed AMEO employability data, including salary patterns by gender and specialization; explored EV pricing and geography with interactive Plotly charts and choropleth maps.

End-of-Studies Internship (Conversational AI) | VERMEG Feb 2024 – Jul 2024

- Built an intent-matching system on Sentence Transformer embeddings (all-mpnet-base-v2), with a fallback to a local Gemma 2 2B model served through Ollama.
- Added multimodal features: PDF Q&A (LangChain, FAISS, PyPDF) and speech processing (SpeechRecognition, gTTS).
- Built the FastAPI/MongoDB backend with JWT authentication and a dialogue builder for global and user-specific intents and text/file responses; streamed replies over WebSockets to a React interface.

Backend Engineering Intern | Proxym Jul 2023 – Aug 2023

- Architected Spring Boot REST APIs for a healthcare app (Nutri), enabling patient management, CRUD, and file uploads.

Data Science Intern | Innomatics Research Labs Feb 2023 – May 2023

- Developed ML predictive models for hardware pricing and analyzed Domino's delivery-time performance data.
- Deployed a Flask URL shortener and a Streamlit data classification application.
- Engineered a rice-disease CNN classifier using OpenCV, utilizing MLflow for experiment tracking.

Frontend Engineering Intern | Bayoudh Jul 2022 – Aug 2022

- Engineered the React user interface for an enterprise MERN-stack inventory management platform.

TECHNICAL SKILLS

AI & LLM Engineering: LangGraph, LangChain, Hybrid RAG, MCP, Gemini/OpenAI APIs, Ollama, LangSmith, RAGAS

Data Science & ML: scikit-learn, pandas, NumPy, TensorFlow, OpenCV, MLflow, Predictive Modeling, Streamlit

Backend & Web: Python, FastAPI, REST APIs, WebSockets, React, Flask

Data & Infrastructure: SQL, MongoDB, Qdrant, FAISS, Docker, Git

SELECTED AI PROJECTS

AI Engineering Lab | Agentic RAG Catalog Assistant LangGraph, Qdrant, FastAPI, React

- Built a LangGraph agent using Gemini and MCP tools for catalog and review search; Qdrant catalog retrieval fuses dense Gemini embeddings with BM25 via reciprocal-rank fusion (RRF).
- Stored graph checkpoints in PostgreSQL, evaluated retrieval with RAGAS, and traced runs in LangSmith.

ResolveIQ | ML-Assisted Triage Engine scikit-learn, LangGraph, FastAPI, Qdrant

- Built a score-first LangGraph workflow on public NYC 311 data (235k+ records), using a frozen Logistic Regression model to predict delay risk, served via FastAPI and a React interface.
- Added optional Qdrant retrieval of semantically similar historical NYC 311 heat and hot-water complaints using Gemini embeddings, with optional borough filtering; traced runs in LangSmith and ran synthetic smoke tests.

EDUCATION

Engineering Degree in Computer Engineering, Data Science Option | ESPRIM 2019 – 2024

CERTIFICATIONS

- LLM Agents MOOC** — UC Berkeley / Berkeley-Haas (2024)
- 365 Data Science:** OpenAI & LangChain Chat Apps (2024); Deep Learning with TensorFlow 2 (2023)